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Best practices for photovoltaic performance loss rate calculations

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E-mail: sascha.lindig@eurac.edu**Keywords:** photovoltaics, photovoltaic performance, degradation models, performance loss rate, photovoltaic ageing

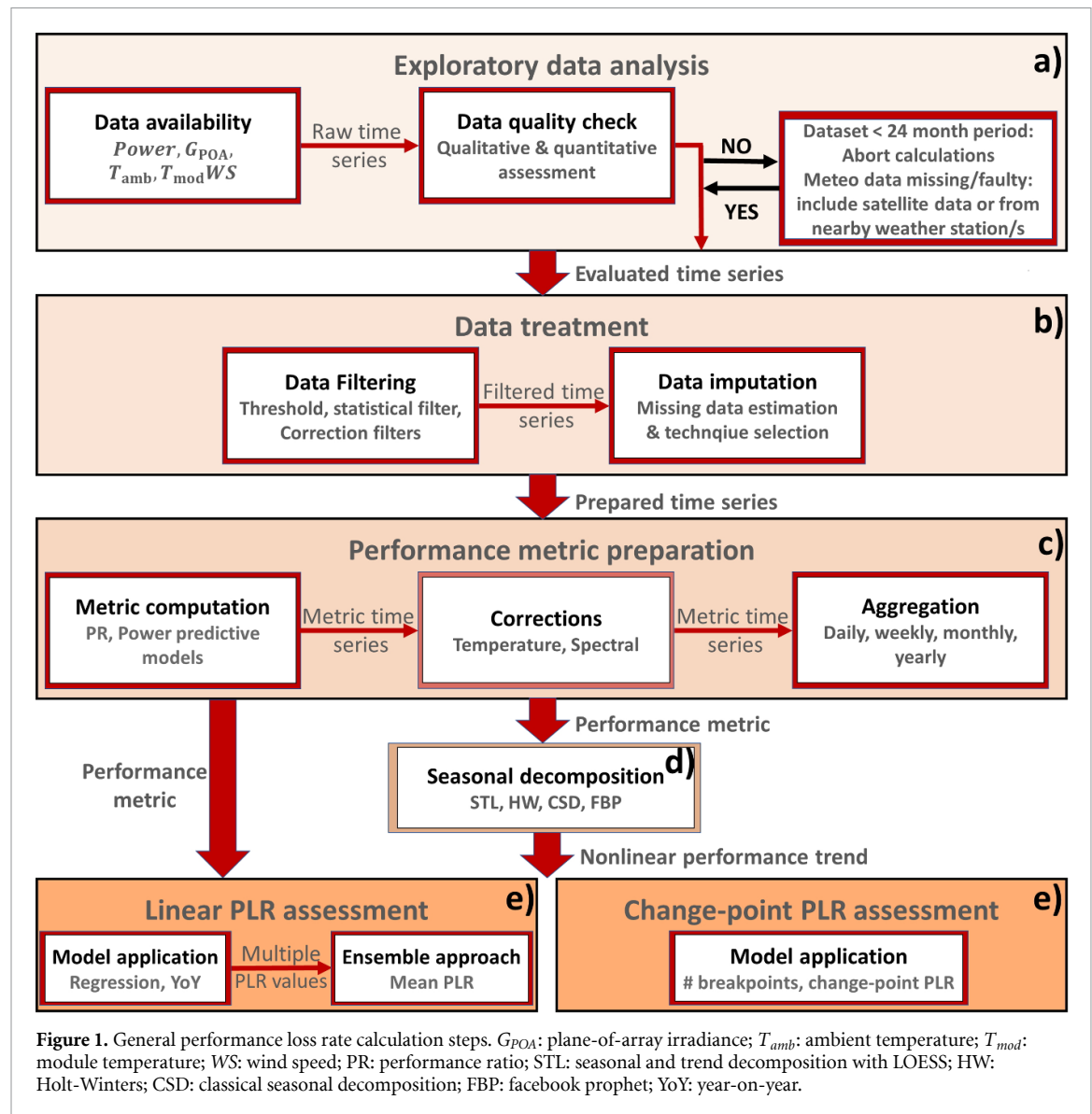
Abstract

The performance loss rate (PLR) is a vital parameter for the time-dependent assessment of photovoltaic (PV) system performance and health state. Although this metric can be calculated in a relatively straightforward manner, it is challenging to achieve accurate and reproducible results with low uncertainty. Furthermore, the temporal evolution of PV system performance is usually nonlinear, but in many cases a linear evaluation is preferred as it simplifies the assessment and it is easier to evaluate. As such, the search for a robust and reproducible calculation methodology providing reliable linear PLR values across different types of systems and conditions has been the focus of many research activities in recent years. In this paper, the determination of PV system PLR using different pipelines and approaches is critically evaluated and recommendations for best practices are given. As nonlinear PLR assessments are fairly new, there is no consent on how to calculate reliable values. Several promising nonlinear approaches have been developed recently and are presented as tools to evaluate the PV system performance in great detail. Furthermore, challenges are discussed with respect to the PLR calculation but also opportunities for differentiating individual performance losses from a generic PLR value having the potential of enabling actionable insights for maintenance.

1. Introduction

The mass deployment of photovoltaic (PV) systems requires efficient and cost-effective operation and maintenance (O&M) approaches worldwide. This includes the reliable assessment of certain key performance indicators (KPI) such as the energy yield, performance ratio (PR), performance index (PI), availability and performance loss rate (PLR). While the definition and computation of parameters such as the energy yield, PI and PR are clearly defined and standardized in the International Electrotechnical Commission (IEC) 61 724-1:2021 [1], there are still misconceptions with respect to the PLR definition and best practices for its calculation. Nevertheless, the reliable calculation of PLR is of extreme importance for a thorough performance assessment and it is the best indicator for a quantitative assessment of the temporal reduction in performance of PV systems.

The PLR can be calculated on both module and system level depending on the monitoring configuration. In this work, the focus is on system PLR, which is more common, since module level assessments require measurement data on module level. The PLR is described in units of %/a, or %/year. On system level, PLR incorporates both reversible and irreversible performance losses such as PV module degradation, soiling, snow, wiring losses, balance of system (BOS) degradation, etc. In the literature, the PLR is often used interchangeably with the term degradation rate (Rd), which is a different metric that represents only the irreversible material degradation of PV modules. This distinction is very important as the degradation rate is only one part of the PLR and the breakdown of the PLR into its components will be the focus of future research to better comprehend time-dependent PV system and module performance. A loss in performance should be denoted negatively, whereas performance gains have a positive PLR value. Furthermore, a distinction can be made between relative and absolute PLR depending if you divide the PLR by the initial metric or not.



PLR calculations should follow an order of pre-defined calculation steps including: (a) input data quality assessment, (b) data cleaning and filtering, (c) performance metric selection together with corrections and aggregation, (d) possible time series feature corrections and (e) the actual PLR calculation using statistical models. A step-by-step guide is shown in figure 1.

Although seemingly simple, the PLR estimation is nontrivial because the selection of each individual step as well as the interplay between these steps determines the final PLR value. Numerous studies have been conducted in order to find the optimal pathway for calculating PLR, especially focusing on data handling, performance metric preparation and application of a statistical model. In this context, Jordan *et al* took the initial version of the year-on-year (YoY) approach of Sunpower [2] as a basis, further developed and refined it [3, 4] and bound it into the RdTools Python package [5]. Furthermore, both Jordan *et al* as well as Phinikarides *et al* conducted studies that showed large deviations in calculated PLR values using preferred calculation methods of different analysts [6–8]. Lindig *et al* carried out in-depth studies looking at the impact individual calculation steps have on the final outcome. Thereby, they rated various statistical models while keeping the calculation steps (a)–(c) constant [9] and also studied the impact of different filters on the performance time series of a PV system [10].

The majority of performance metrics used in PLR pipelines require irradiance as input. Thereby, ground-measured irradiance time series are the preferred choice but often they are not available or of questionable quality due to neglected maintenance and/or calibration of the sensors. Data quality assessment of the irradiance measurements is therefore indispensable, including possible sensor drifts. An alternative is the usage of satellite-based irradiance data or ground-measurements from nearby weather stations, but these options may introduce bias in the performance metric and consequently the PLR estimation. Other

interesting alternatives exploiting clear-sky methods and conditions have been explored and developed by Jordan *et al* [4], Meyers *et al* [11, 12] as well as Ascencio-Vasquez *et al* [13]. The advantage of such methods is that they are independent of any irradiance source. However, these approaches also exhibit shortcomings such as the inability to account for long-term irradiance trends such as solar brightening/dimming effects, seen in recent year across large parts of the globe [14, 15].

All these advancements in the field helped tremendously to forge best practices in the calculation of PLR. Ultimately, this led to a large benchmarking study within the International Energy Agency (IEA) Photovoltaic Power System Program (PVPS) Task 13 [16] where the question of the most reliable calculation approach for linear PLR was critically investigated. It was shown that a ‘perfect’ unique combination of filtering, performance metric and statistical calculation method does not exist because the individual calculation steps form complex interactions with the characteristics of the PV system performance datasets. In this study, guidelines for the PLR calculation and the complexities of the topic at hand are presented. Every calculation step is discussed individually and suggestions for more reliable assessments are provided, such as how to carry out a data quality assessment, when and how to use data imputation techniques to provide unbiased results or what to do if no ground-measured irradiance data are available. Furthermore, the relatively new field of nonlinear PLR assessment is discussed and the implications for its usage are shown.

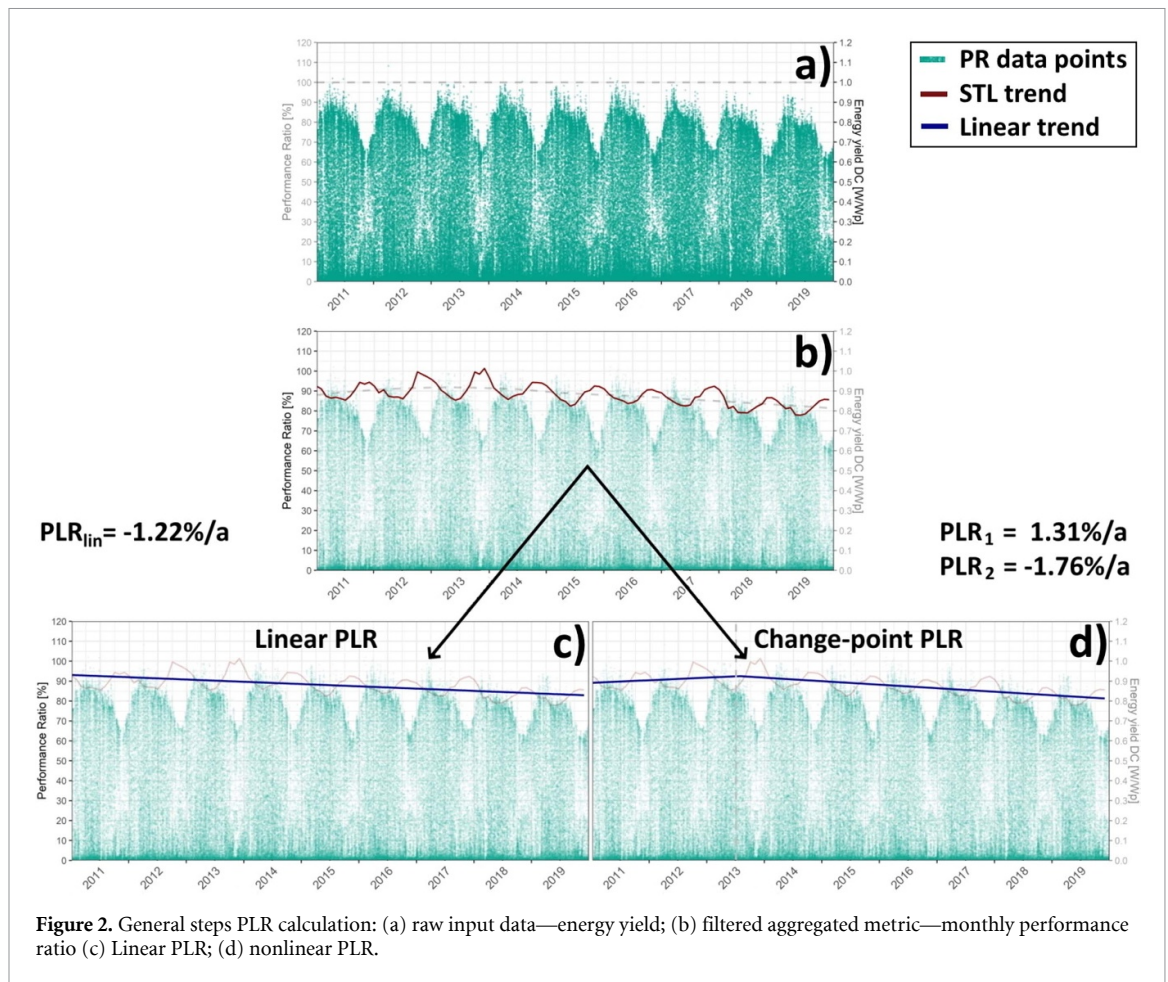
Section 2 differentiates linear from nonlinear PLR and provides a background and a brief literature review on nonlinear PLRs. Section 3 provides an overview on the theory and calculation steps involved including advancements in best practices for the calculation of linear PLR. This part is based on a large benchmarking study carried out in the IEA PVPS Task 13 [16]. In section 3.4.2, new approaches for calculating nonlinear PLR are introduced. The nonlinear evaluation opens up new possibilities for incorporating PLR not just in lifetime yield or overall loss trend assessments, but also as a KPI in O&M strategies to detect performance anomalies and in technical and financial due diligence for the change over in O&M operators or in the secondary market. Section 4 discusses the challenges and opportunities in PLR estimation. The paper closes with a short summary and conclusions in section 5.

2. Nonlinear performance loss rates

A common assumption when calculating PLR is of linear behavior. Performance data show that such an assumption may not be realistic in most cases. This is due to reversible and irreversible performance losses that may occur at different stages and duration during a PV plant’s lifetime causing nonlinear trends. Although such behaviors can have a significant impact on a project’s economics [4, 17, 18], only recently methodologies for estimating nonlinear Rd and PLR were introduced based on change-point analysis [19–23].

Theristis *et al* [19] introduced Facebook Prophet (FBP) and compared it against conventional models which assume linearity finding LCOE differences up to 6%. Lindig *et al* [22] introduced a multi-step approach based on piecewise regression, which was then applied on real system data from Bolzano, Italy. Since a variety of open-source change-point libraries is available in both Python and R, a comparative analysis of six different libraries was conducted recently [23] in order to find the most appropriate change-point models for PV Rd and PLR applications. This analysis found that piecewise regression and FBP are the best performers with respect to detecting and quantifying nonlinear degradation. Other libraries, such as the ‘Bayesian estimation of abrupt change, seasonality and trend’ (RBeast) [24], breakpoint (bkp) [25] and ‘Sequential and batch change detection using parametric and nonparametric methods’ (cpm) [26] could still be used in PV monitoring applications although they exhibited higher errors (mean absolute errors within 0.3%/year). All open-source change-point libraries require knowledge of the number of change-points and therefore, customization is required to automatically detect statistical changes in PV performance time series. Typically, once these n changes are detected and verified as change-points, the time series is divided in $n+1$ different segments and linear regression is applied to extract $n+1$ PLR values instead of a single one. More information on these libraries and automatic change-point extraction are discussed in section 3.4.2.

As described here, PLR can be calculated in a linear or sublinear fashion, depending on the calculation approach and purpose of the study. While a linear assessment is somewhat the standardized method, a sublinear assessment allows to evaluate a PV system’s performance in greater detail. Figure 2 depicts a generalized and simplified PLR calculation pipeline of performance data of a mono-crystalline silicon PV system installed at the multitechnological PV test field at the Airport Bolzano Dolomiti (ABD) in the Italian Alps. Both, the linear and nonlinear (or change-point) PLR procedures and values are shown. The detailed calculation steps are discussed in section 3. The steps (a) and (b) in figure 2 respectively (a)–(d) in figure 1 are identical regardless if a linear or a nonlinear PLR assessment is performed.



After preparing the input data, the nonlinear performance metric (i.e. monthly PR in figure 2) is computed and the PLR is then calculated. In figure 2, it is visible that this particular system is operating with a moderately consistent performance behaviour. Nevertheless, the change-point PLR approach allows the detection of an seemingly initial performance gain for the first two and a half years of operation before the PLR drops to a value of $PLR_2 = -1.76\%/year$. The detected ‘gain’ is believed to be caused by an unusual sunny and cold winter in 2013/2014, which provided optimal performance conditions for crystalline silicon modules. The selected example might not be an extreme case of individual performance impairing faults to be detected by a change-point PLR algorithm, but it clearly shows the advanced evaluation possibilities given by a more detailed PLR assessment such as the change-point detection approaches.

3. Commonly used pipelines

In this section, commonly used approaches for calculating PLR are presented, divided into the main calculation steps presented in figure 1. It is very important to carry out all described steps in the right order to (1) prepare and assess the input dataset, (2) clean the data thoroughly, (3) compute a suitable performance metric and (4) calculate the PLR using reliable calculation approaches. Based on the investigations carried out in [16], the best strategy to calculate linear PLR is by following the ensemble approach. Thereby, a number of methods (as presented here) is used to calculate individual PLR values, divide the results into statistically similar approaches and outliers, and then calculate the mean over the inlier values to receive a ensemble PLR. The R package PVplr [27] has been developed to follow this calculation approach.

In order to divide the tested approaches into being statistically similar and outlying, the 95% confidence interval for each individual approach is calculated as uncertainty and approaches with overlapping confidence intervals are to be considered as statistically similar. The ones with confidence intervals out of the range of overlapping intervals are categorized as outliers and should be removed before computing the ensemble PLR. An example for the usage of the ensemble approach can be seen in figure 3. Here, individual PLR of a poly-crystalline PV system have been calculated using 27 different approaches. After the outlier categorization, the ensemble PLR is equal to $\overline{PLR} = -0.88\%/year$.

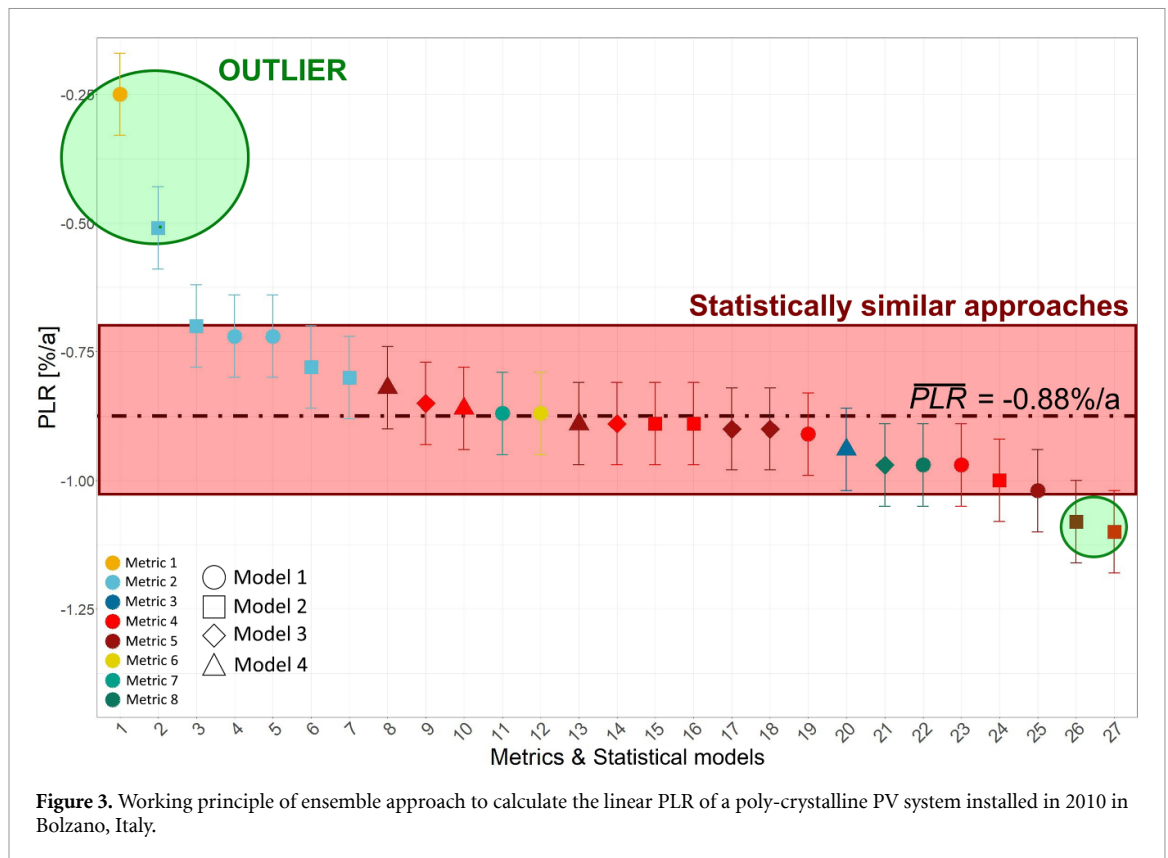


Figure 3. Working principle of ensemble approach to calculate the linear PLR of a poly-crystalline PV system installed in 2010 in Bolzano, Italy.

3.1. Input data quality

The first block in the calculation pipeline, shown in figure 1, includes reading the input data and assessing its quality during the initial exploratory data analysis. It is strongly recommended to thoroughly perform this analysis to ensure a smooth implementation of the following calculation steps. Thereby, performance data from the respective PV system together with ground measurements of climate sensors (mainly irradiance, temperature, wind speed) or climate data from other sources are prepared. While joining performance and climate data from different sources, it is important to ensure that the timestamps are aligned. This issue may seem trivial but a shift between the measurement signals impacts inevitably the consecutive steps and thereby falsifies the final result.

Additionally, qualitative and quantitative data analyses are indispensable to evaluate which datasets are robust to be analyzed and which might fail at certain calculation steps/types. An example is the effect of missing data rates within the dataset [28, 29], which is further discussed in section 3.2.2. Qualitative data analysis can be performed by visualizing the measurement signals over time or versus one another [10], exemplarily shown for performance data of a PV system installed in Bolzano, Italy in figure 4. Here, strong deviations from the norm, such as outliers, missing data, inverter clipping, sensor alignment issues or other data related issues, can be detected. However, the system data shown in figure 4 can visually be considered of very high quality without any apparent issues.

Figures 4(a) and (b) show normalized energy and power time series, respectively. Normalization is performed by dividing a given datapoint with the nominal power (P_{nom}) of the system. Power is measured as instant values or as the average of single datapoints in sub-hourly resolution and is given in Watts [W]. In contrast, energy values are aggregated power measurements given in Watt-hours [Wh]. The same relation holds for irradiance [W] and irradiation [Wh]. Figure 4(c) shows the measured power over the plane-of-array (POA) irradiance. As power and irradiance have a nearly linear relationship, this plot helps to assess the irradiance data quality in terms of sensor alignment, calibration drift and similar issues based on abnormal behaviour. The outlier cloud at the bottom of the graph is caused by row-to-row shading appearing in winter months. Figure 4(d) shows the Performance Ratio (PR), which is calculated according to equation (1). When studying PV performance, both DC or AC related performance metrics can be considered. When using DC metrics, possible influences due to inverter degradation or performance impairing issues (e.g. temperature derating) can be eliminated, which is in many cases advisable. The PR is a unit-less parameter, and it is usually used as a quality measure by comparing the produced energy with the expected energy of a PV plant and is thereby a recommended metric for performance inter-comparisons of different PV systems.

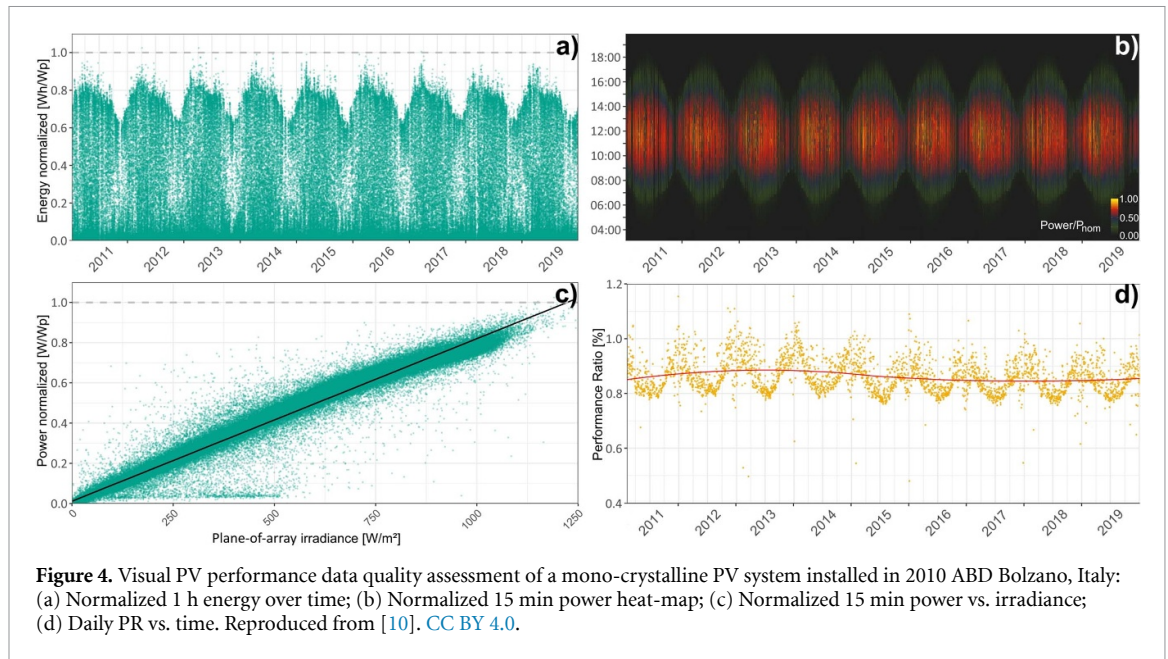


Table 1. Data quality grading scheme: outliers (include the impact of clouds, and anomalous datapoints) and missing data (excluding nighttime) are assessed in the highest data resolution, and the longest gap in days. The dataset length needs to be greater than two years for a P (Pass) grade, otherwise it is graded with F (Fail).

Letter Grade	Outliers (%)	Missing percentage (%)	Longest Gap (days)
A	Below 10	Below 10	Below 15
B	10–20	10–25	15–30
C	20–30	25–40	30–90
D	Above 30	Above 40	Above 90
P/F	Time series $\geq$ 24 months = > Pass		

Although a visual and qualitative data assessment does give a good initial overview on the state of the input data, a quantitative assessment is required to evaluate a dataset's quality in greater depth and also to be able to inter-compare the different dataset characteristics. Within the IEA PVPS Task 13, a new grading scheme has been developed [16, 30] for assessing data quality with respect to three areas: the amount of outliers, missing values and the largest gap within a dataset. Furthermore, a simple Pass/Fail criterion is introduced, determining if at least two full years of data are available (accepting imputed values under certain conditions) as this is the theoretical minimum for PLR calculations, as the PLR describes an annual performance change. The grading criteria can be seen in table 1.

The data grading scheme is a tool to statistically evaluate a datasets quality. The scheme has been developed/adapted for application to PV system time series and it is based on an approach used for building electricity time series [31].

Outliers are usually defined as points greater than ± 1.5 times the interquartile range (IQR) [28]. Software packages such as `tsoutlier` [32] or `kats` [33] allow for a more advanced outlier detection through the usage of seasonal decomposition. Filtering methods discussed in section 3.2 are applied to remove outliers in PV performance datasets. Missing datapoints as well as longer data gaps are other sources of errors, which may bias the PLR estimation. Missing data, especially over longer periods, are difficult to correct for or impute and mitigate their impact.

Finally, the time series length is checked together with the availability of ground-measured climate data. At least 24 months between the first and last month are required for a year to year performance loss assessment. Depending on the completeness of data, imputation techniques may be used to fill the time series. For example, the dataset used in the figures 2 and 4 has a data quality grading of BAAP and can thereby be considered as being of high quality. In section 3.2.2, the impact of missing data rates and imputation strategies are discussed.

3.2. Data filtering and imputation

After reading the input time series and assessing its quality, the data are prepared for the subsequent PLR calculation steps via filtering and, if needed, data imputation.

3.2.1. Data filtering

Data filtering is one of the most crucial steps of data preparation. Filters are applied to remove data which deviate from the known relationship between the output of a PV system on one hand and incident irradiation and temperature on the other. A special case appears if this relationship is affected due to maintenance actions that temporarily reduce the power output. Here it depends on the availability of maintenance tickets if these time periods can be detected and kept in the dataset. In the following, general filtering concepts are presented. Filtering for PLR calculation studies can be divided into threshold and statistical filters, clear-sky filtering and others; e.g. filtering to correct for inverter curtailment or shading, soiling and snow events.

A first assessment of the impact different filters have on the overall PLR was conducted by Jordan *et al* [34]. Another study highlighting the importance of filtering on PLR calculations was carried out by Paudyal *et al* [35]. Next to the work mentioned above, a study from Livera *et al* [36] focuses on the question which filters should be applied for PLR calculations and/or by how much whereas oftentimes they seem to be selected arbitrarily. Filtering criteria have to be selected carefully because they may introduce significant bias on the PLR results, as shown in [16].

3.2.1.1. Irradiance threshold

Irradiance thresholds are a standard set of filters that are commonly applied on PV performance time series. Initially, as the lower cutoff, high irradiance thresholds of around 800 W m^{-2} were preferred to focus on optimal performance conditions closer to STC. Lately, lower thresholds of $100\text{--}200 \text{ W m}^{-2}$ have gained acceptance to avoid extensive data removal and to capture a wider performance spectrum, providing a more complete picture of PV performance. The lower threshold is primarily set to remove nighttime values and low irradiance periods. High thresholds, usually set at around 1200 W m^{-2} , are used to remove outliers caused by cloud reflections and measurement errors. Furthermore, the upper threshold depends on the monitoring system used. Data read directly from the inverter have lower uncertainties at high irradiance [37].

3.2.1.2. Power threshold

Power thresholds are usually set to remove outages and measurement errors. Power threshold filters are closely related to irradiance thresholds as they are oftentimes affecting the same datapoints. Power thresholds need to be tailored based on the PV system under investigation due to differing installed capacities. A recommended way to account for differing nominal power values is by setting thresholds as a percentage of the nominal power or the maximum detected power.

3.2.1.3. Statistical filter

Statistical filters are used to detect and remove outliers deviating from the power-irradiance relationship of PV systems, which are not captured by any threshold filter. In this case, performance metrics such as the performance ratio or empirical power models are computed and deviation intervals based on standard deviations or interquartile ranges around the performance metric are set. This filter is useful for capturing reduced performance that is not related to degradation or erroneous measurement events such as shading of the irradiance sensor while the PV system is exposed to the Sun or vice versa, increased soiling instances or similar effects.

3.2.1.4. Clear-sky

This type of filter is based on clear-sky models, which model irradiance conditions with no cloud cover at the location of interest. Clear-sky filters are used to decrease the impact of inconsistent shading on PV systems. This includes partial shading of parts/strings of a PV plant or shading of only the irradiance sensor or the plant. Furthermore, clear-sky filters are useful for detecting irradiance sensor drift by comparing it to modeled irradiance values under clear-sky conditions. Commonly used clear-sky models are available in `pvl1b-python` [38], which are also used in the `RdTools` [39] PLR calculation pipeline. More specifically, the clear-sky index (CSI) method is used in `RdTools` and identifies clear-sky periods by providing a ratio between the modeled and measured irradiance signal. By default all sensor-based irradiance periods within 15% of the modeled clear-sky irradiance are flagged as clear-sky conditions.

3.2.1.5. Inverter saturation and curtailment

Inverter saturation appears when the DC power output of a PV system exceeds the rated AC power output of the inverter. The reason is the selected inverter loading ratio (ILR), which describes the DC-AC capacity ratio of PV systems, resulting in clipping. Especially in large scale PV systems, ILRs of 1.13–1.30 are observed [40]. Benefits of higher selected ILRs include increased capacity factors as well as stabilized power outputs. Inverter saturation is detectable in PV power trends by flat plateaus at the peak, usually under high irradiance conditions. Untreated saturation events may falsify a PLR analysis by including power values which are no

longer a function of the prevailing weather conditions. Curtailment on the other hand is defined by the inability to inject the energy produced by a power plant into the distribution network, which is controlled by the electricity authority. Therefore, it is recommended to filter out such events by identifying the saturation limit and putting a threshold of 99% around it.

3.2.1.6. Shading, soiling and snow

Shading, soiling and snow are effects that can trigger power losses across parts of a PV plant or even the entire PV plant and they are difficult to detect from PV power time series. Snow coverage or strong soiling is sometimes reported in systems logs, although this may not be a reliable/consistent source. If the local irradiance sensors are calibrated and maintained, it is possible to identify such events using irradiance corrected performance KPIs (e.g. PR). During such events, the power conversion is typically lower compared to 'standard' operating conditions. Therefore, the most reliable strategy for removing such instances is by using statistical performance filters or clustering approaches of power and irradiance trends [4, 27]. Although it is common to remove such instances as they are usually time limited and do not represent actual PV system performance, caution is recommended when large periods of such effects exist; excessive filtering may introduce significant bias in the estimated PLR [36].

3.2.2. Data imputation

As mentioned earlier, missing and outlying data, which can be caused by outages and/or erroneous measurements, are common in PV datasets. Although these can significantly affect the accuracy of the applied statistical methods, the majority of PLR pipelines treat missing and invalid measurements by filtering them out. However, it was shown that when missing data rates exceed 10% [36, 41], the PLR estimate deviates significantly from its true value. Furthermore, the missing data rate limit depends also on the length of the dataset meaning that shorter time series will require the highest data availability possible. This was investigated by Romero-Fiances *et al* [41] where 3–5 y of data were found to be the minimum required for Rd or PLR deviations within $\pm 0.2\%$ /year, given a data availability of 100%. Longer time series (e.g. of 10 years), were found to be robust to datasets with missing data rates of up to 10%; higher rates resulted in PLR deviations in excess of $\pm 0.6\%$ /year, which is significant. Therefore, data inference techniques are recommended for reducing bias in PLR estimations.

A complete and justifiable pipeline for PV data processing, quality verification and reconstruction was recently presented by Livera *et al* [36]. This methodology includes data statistics, consistency examination, filtering, detection of erroneous measurements and the application of models for inferring gaps caused by missing data and erroneous values in PV performance time series. Both physical and statistical inference techniques were examined and their ability to reconstruct PV operational datasets was quantified. It was shown that power (Sandia Array Performance Model, PVWatts, etc) and temperature (Sandia Module Temperature Model, Faiman Model, etc) models proved to be adequate for inferring missing values when missing data rates are greater than 10% whereas statistical methods such as the predictive mean matching are robust for inferring irradiance measurements.

3.3. Performance metrics

Performance metrics are calculated by normalizing the power time series. These types of metrics are preferred to be used for PLR calculation pipelines over raw power time series, as they contain the underlying PV system performance more reliably with less noise. PLR calculation approaches using measured power with the aid of strict irradiance and temperature filters have been used but are not recommended as they remove a large amount of datapoints [16].

3.3.1. Performance ratio

The performance ratio (PR) is a unit-less parameter which describes the relation between the energy yield of a PV system (DC-side—array yield; AC-side—final yield) and the reference yield, which is the yield the system is expected to generate based on local climate conditions. It is calculated by [1] :

$$PR_{DC(AC)} = \frac{Y_{a(fin)}}{Y_{ref}} = \frac{E_{DC(AC)}/P_{nom}}{H_{POA}/G_{STC}}. \quad (1)$$

Yields Y are ratios of measured values of energy E or irradiation H to values obtained under Standard Test Conditions [42] (STC: solar irradiance = 1000 W m^{-2} , cell temperature = 25°C , wind speed = 1 m s^{-1} , air mass = 1.5). Here, E is the energy produced by the PV system in a defined time period t and P_{nom} is the nominal power of the PV plant under STC conditions. H_{POA} is the plane-of-array irradiation and G_{STC} is the reference irradiance of 1000 W m^{-2} .

The PR may be corrected for temperature ($PR_{T_{corr}}$) to reduce seasonal variations of the PR due to the temperature dependency of the used PV cell. Thereby, temperature coefficients are provided by the manufacturers or can be calculated using measured time series data. The correction should be performed according to IEC standard procedures [1]. It has to be noted that, although the correction lowers seasonal fluctuations, it does not diminish them entirely due to other effects such as the impact on performance of the angle of incidence and the solar spectrum of the incoming light.

3.3.2. Performance index

Performance index (PI) is a metric that describes the ratio between the measured energy of a PV system and the predicted energy using a power predictive model. They are usually build for certain time periods, ranging from days to years, based on the relationship between weather data and PV system output. The inputs of power prediction models are measured weather data while the outputs include efficiency and/or maximum power. PI values are close to 1 for PV systems without extraordinary performance impairing issues and should not be subject to seasonal variations based on temperature variations, unlike the PR. PV performance models which were found to be used in PLR evaluation applications include the PVGIS, PVUSA, XbX and the PVWatts model. The first three require training to fit the coefficients and the models are built over defined time periods to include the changing PV performance [43–45]. On the other hand, PVWatts is a simple power model using spec sheet information and does not require any fitting [46]. Despite its simplicity, PVWatts offers adequate accuracy, which makes it the most preferable and popular model for PI calculation in PLR applications [4, 39]. More details on the models can be found in French *et al* [16].

3.3.3. Others metrics

Apart from the PR and PI metrics, other performance metrics were reported in literature for PLR calculations; e.g. the energy yield and normalized power. Normalized power is calculated by dividing the measured power of a PV system by the nominal power of the PV plant P_{nom} under STC conditions. As both, the energy yield and normalized power, do not have any corrections terms in regards to irradiance or temperature, they are usually applied in combination with very strict filters such as narrow irradiance bins (e.g. ± 50 to $\pm 100 \text{ W m}^{-2}$), depending on data availability. These intervals are set around STC, nominal operating cell temperature (NOCT) or nominal module operating temperature (NMOT) conditions. The application of such narrow irradiance filters is not recommended because they filter out large parts of the data. Furthermore, the remaining data are representative of a narrow range of operating conditions, which may not be representative of the PV plant.

3.4. Statistical models

After preparing the performance metric and aggregating the time series to a desired time step (usually days, weeks or months), the PLR is calculated using a statistical model. Thereby, a distinction between linear and change-point algorithms is made. A linearized performance loss representation is still the common approach but change-point PLR approaches gain importance due to the possibility of describing a PV systems temporal change in performance in greater detail. Initially, linear regression was often the model of choice. In recent years, many different statistical models were adapted or developed, which oftentimes show better performance as compared to linear regression. In this section, the most common statistical models are discussed.

3.4.1. Linear performance loss rates

Linear PLR methods can generally be divided into regression methods and methods based on a loss rate distribution. As such, regression based methods are commonly used in conjunction with seasonal decomposition approaches.

3.4.1.1. Linear regression (LR)

Least-squares linear regression is a commonly used approach due to its simplicity. The fitted trend line is given by:

$$\hat{y} = \beta_1 t + \beta_0. \quad (2)$$

β_1 is the gradient and β_0 the intercept with the y -axis. A least-squares regression fits linear trend lines to the nonlinear performance metric time series and minimizes the sum of the squared distance between datapoints of the metric and the trend line.

An alternative to classical least-square regression is robust regression which is less sensitive to outliers by adding weights to the regression function.

3.4.1.2. Seasonal trend decomposition using loess (STL)

STL is a seasonal decomposition algorithm which separates the nonlinear trend of a time series from its seasonal and residual component in an additive manner using locally weighted, non-parametric regression [47, 48]. After decomposition, LR is applied to the nonlinear trend line to get a regression representation of the performance evolution of the PV system. Here, the gradient is multiplied by a factor to present yearly values (12 for m, 365 for d etc) to compute the final PLR. According to [9], this method showed reliable results for time series with seasonal behaviour and high data quality. STL functions are available both in R and Python [49–51].

3.4.1.3. Year-on-Year (YoY)

This method was first applied for PLR determination purposes by Hasselbrink [2] and is extensively used through its inclusion in the Python RdTools package as well as the PVplr R package [27, 39]. It is based on a loss rate distribution rather than a single value. Thereby, a performance metric datapoint (resolution depends on the selected aggregation step and can be hours, days, weeks or months) in one calendar year is compared to the value at the same time in the subsequent year and the gradient is stored. This is done across all available datapoints, and years, and the median of the gradients across all datapoints determines the PLR. The PLR computed with the YoY method is usually normalized to the first-year's median [4, 39].

3.4.1.4. Other methods

Apart from the three primary approaches listed above, other methods have been tested or developed, which are introduced below. Just like in the case of STL, models based on data decomposition are followed by linear regression to linearize the decomposed trend line.

Classical seasonal decomposition (CSD) is a decomposition method similar to STL but based on a centered moving average [48]. The selected seasonal period depends on the data resolution and it is usually set to 12 for monthly data. In that case six months at the beginning and six at the end of the observation period are not included in the smoothed time series. The exclusion of data points is due to the nature of the moving average and that is why CSD is not recommended for short time series [9].

Holt-Winters (HW) seasonal modeling is a rather complex decomposition approach based on a weighted averaging algorithm to determine the trend line. It is composed of a forecast equation and three smoothing equations including a level, a seasonal and a slope component. It can either be additive or multiplicative. For PV time series, the additive method should be used as seasonal variations are expected to be constant throughout the time of observation. It can be used both for smoothing and forecasting performance time series of PV systems [48, 52].

Statistical-clearsky-fitting (SCSF) fits a constrained, non-parametric clear-sky model to the power time series of a PV system and detects a year-over-year percentage change as PLR similar to the YoY approach. The input of the model is a PV system's daily energy profile and therefore, it does not require any additional data such as meta- nor environmental data [11, 12].

The VAR Method was developed by Meftah *et al* [53] specifically for PLR assessments. It is based on fitting regression models of power variations with respect to environmental variations (irradiance and ambient temperature). The basic idea is to build a correlation model between annual variations of output power with respect to annual variations of environment. The VAR method provides yearly PLR rates and by averaging the annual values, the PLR of a PV system is computed. A regression according to the following relation is fitted:

$$\Delta Power = f(\Delta G, \Delta T_{amb}) + d. \quad (3)$$

If f is accurate enough, d is the variation of power independent of environmental changes, and interpreted as a performance loss.

3.4.2. Methods for nonlinear performance loss rate calculation

Pipelines for nonlinear PLR estimation are similar to the ones applied for linear PLR. The only difference in nonlinear PLR extraction is that once the time series is prepared, normalized and aggregated (see figures 1(a)–(c)), a change-point model is applied to automatically detect the number and location(s) of change-points based on changes in the statistical properties (e.g. slope, mean, variance, etc). Furthermore some change-point models are benefited by trend decomposition (figure 1(d)), which deseasonalizes the signal, whereas others perform well by directly extracting change-points from the normalized and aggregated time series.

3.4.2.1. Facebook Prophet

(FBP) is a modular regression model. It has an additive structure including a trend parameter, seasonality, a holiday term (ignored in PLR calculations) and an error parameter. The yearly seasonality option of the FBP model is set to TRUE treating monthly behaviour as PV power time series are expected to show monthly seasonality patterns. The trend is fit using a piecewise linear and a saturating growth model [54]. The FBP model is well suited for nonlinear PLR estimation because of its ability to perform change-point analysis [55] on the time series trend either automatically or by modifying a number of parameters. In order to use FBP for PLR analyses, the flexibility of the extracted trend, number of potential change-points, and range has to be adjusted according to Theristis *et al* [19, 20]. One way to enable automated change-point detection is by utilizing the `signal.find_peaks` function from SciPy [56]. Furthermore, if the change-point detection step is ignored, FBP can also be used for linear PLR extraction; i.e. by directly applying linear regression on the extracted trend [16, 30].

3.4.2.2. Piecewise (or segmented) regression

Piecewise (or segmented) regression has been used to estimate nonlinear PLR with STL decomposition [22] and without any decomposition [23]. In the former approach [22], STL was applied to the prepared performance metric (PR or PR_{Tcorr} depending on climate data availability) to deseasonalize the time series. In order to automatically find the number and position(s) of change-point(s) in the performance signal, the separated nonlinear performance trend was analyzed using an adapted version of $R^2_{adjusted}$; a parameter used in multivariate regression to determine the optimal number of predictors for a model, together with the R package `segmented` [57], which fits a regression model with segmented relationships between the performance time series and one or more explanatory variables. The idea is to create n functions with $n - 1$ potential change-points using `segmented`, and to use the aforementioned parameter to find the optimal function by evaluating function accuracy versus function complexity. With respect to the Pythonic piecewise regression, which can be utilized using SciPy's `curve_fit` function [56] one can automate the process of PLR change-point detection by examining the standard deviation returned for the date of each detected change-point; more details are given elsewhere [23].

Although piecewise regression and FBP exhibited similar accuracy when compared against synthetic datasets [23], piecewise regression libraries have the advantage of being much faster than FBP. However, FBP offers additional functionalities such as forecasting and the ability to control the flexibility of the trend opening up opportunities toward potentially breaking down the PLR into individual performance losses.

3.5. Uncertainty estimation

Independently of the calculation approach used, a thorough data quality check combined with a careful selection of filtering approaches are crucial steps in the preparation of a performance time series for PLR calculations. The individual calculation and analysis steps are strongly correlated. This may suggest that the most reliable path towards linear PLR assessments is by using multiple 'filter-metric-method' approaches, and to determine the voted mean PLR as the most reproducible and accurate result. Therefore, the standard error of the population mean across the PLR determination results using 95% confidence intervals is computed to divide the results into being statistically similar (i.e. inliers) or outliers before computing the mean. It is important to establish common uncertainty determination approaches regardless of the calculation approach used to ensure comparable confidence intervals which may or may not overlap.

In order to evaluate the 95% confidence interval of regression-based PLR determination approaches, the variance of the slope of the linear model coefficients are calculated according to [9].

The 95% confidence interval of regression-based PLR determination approaches can be evaluated from the variance of the linear model coefficient's slopes according to [9]:

$$PLR = \frac{12a}{b}, \quad (4)$$

$$u_{PLR} = \sqrt{\left(\frac{12}{b}\right)^2 \times u_a^2 + b^2 \left(\frac{12a}{b^2}\right)^2 \times u_b^2}, \quad (5)$$

a and b are the regression fitting coefficients, $u_{a,b}^2$ the variances of these fitting coefficients and u_{PLR} the standard deviation of PLR. When using this method with decomposed time series the residuals component should be added back into the trend component so that the final time series has the same signal to noise characteristics as a PLR determined without using decomposition. Like this, PLR values based on regression with and without decomposed time series are directly comparable.

The uncertainty of YoY based PLR is indicated with a confidence interval calculated using a Monte-Carlo, or bootstrap re-sampling, of a probability density function of the individual PLR results [39, 58]. When

calculating the PLR using the YoY function provided within the RdTools package [5], the confidence interval can be set manually.

A direct comparison of regression and YoY based uncertainty is difficult because of the varying approaches explained above. A solution to this problem might be the application of bootstrap resampling with replacement to compute the uncertainty, by assessing the standard error of the distributions. For regression-based PLR estimates, the bootstrap is applied to the time series where 65% of the datapoints are randomly selected and the PLR is recalculated for a number of times. Instead, when applying a bootstrap to YoY-based results, the loss rate distribution is resampled instead of the time series. 1000 bootstrap iterations are usually carried out to ensure uncertainty convergence.

4. Challenges in performance loss rate estimations and opportunities

PLR calculation is relatively straightforward, but the pipelines used to extract accurate and reproducible values are rather complex as presented throughout this manuscript. Vague assumptions, high uncertainty and/or inaccurate estimates may result in high financial impacts. As a simplistic example, consider a 30-y project with a 100 MWp plant, $0.05 \text{ \$ kWh}^{-1}$ electricity price and 1100 kWh/kWp/y specific yield, which is typical for Bolzano, Italy. Assume also three different PLR scenarios based on what is commonly used by the industry and the estimates from figure 2; i.e., (a) linear PLR of $-0.5\%/y$, which is a common assumption within the industry, (b) a linear PLR of $-1.22\%/y$ and (c) a nonlinear PLR profile with $PLR_1 = 1.31\%/y$ and $PLR_2 = -1.76\%/y$ (see figure 2). The undiscounted 30-year revenue estimates for these three scenarios would be (a) \\$152.2 million, (b) \\$133.8 million and (c) \\$121 million; this is a maximum of 25.8% difference in revenues. Although this is an example with various simplistic assumptions, it nevertheless signifies the importance of accurate PLR determinations and the need to overcome the main challenges. The detailed understanding of PLR of operating plants will allow us to make better performance predictions for future PV plants in respect to the prevailing climate conditions and PV technology. This in turn will improve long-term yield predictions, thereby reducing related uncertainties to provide a better business case for future PV projects.

4.1. Unavailability of ground-measured climate data

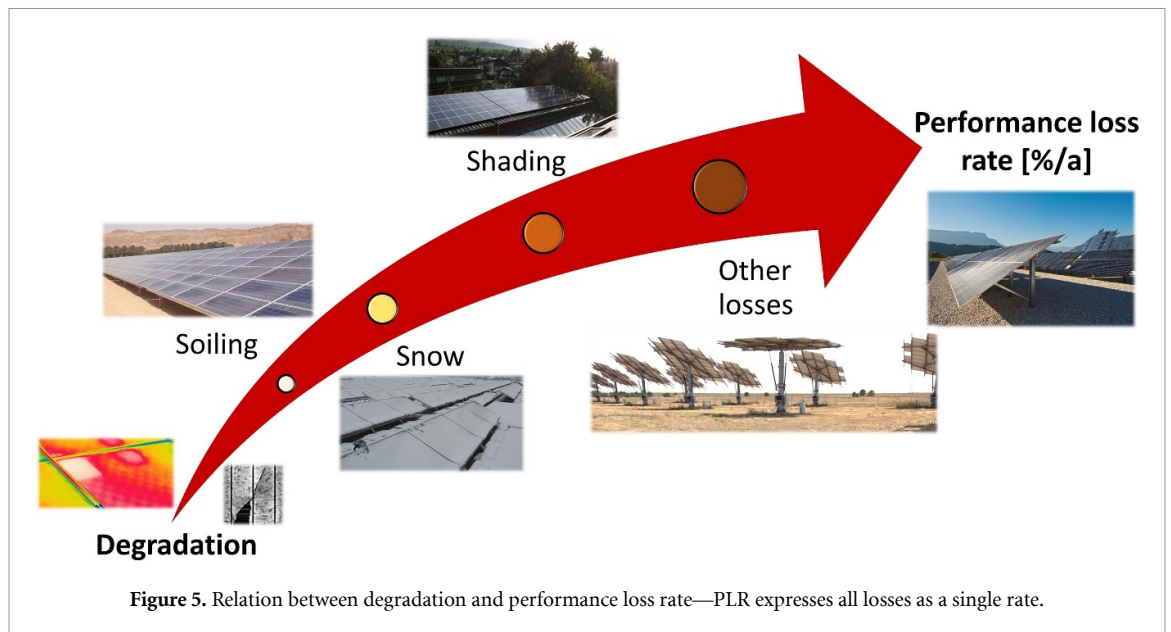
Apart from the discussed challenges of PLR calculations, other issues may occur which add additional complexity and uncertainty in the results. An often occurring and problematic situation is the unavailability of ground-measured climate data, either because no dedicated weather station has been installed or the installed equipment is not cared for to a sufficient level. If the sensors are not regularly cleaned and calibrated, sensor drifts and measurement bias may appear and have a negative impact on the data quality. Different solutions have been developed and tested to overcome this issue. These include the development of models, which do not need meteorological data, the usage of alternative weather data collected within dedicated databases or measured at nearby weather stations, or the application of site adaptation techniques.

4.1.1. Weather-agnostic models

The first option is to apply models which do not rely on weather data such as irradiance or temperature. In theory, all PLR models discussed in 3.4 can be applied to a performance metric only relying on the power time series but the outcome will include many unwanted measurement artifacts and therefore, it is subject to large uncertainties. Interesting representatives of these types of models are the SCSF model [11, 12], discussed in 3.4.1, and the YoY approach using a modeled clear-sky irradiance time series instead of measured irradiance. A clear-sky model assumes consistent sky conditions, thereby possibly ignoring local effects. This in turn can bias PLR values unintentionally especially for shorter time series. Another possible shortcoming of this type of model is that it excludes any short- and long-term variations in the climatic conditions. Solar brightening effects have been observed since the early 1980s to this date, describing a rise in annual solar irradiation on the earth's surface [14]. For instance, Kiefer *et al* [59] observed an average increase in irradiation of $+1.1\%$ per year in Germany.

4.1.2. Databases

Another solution is the usage of weather data coming from online databases. The data may be composed of ground measurements from weather stations or processed satellite images. Data are available from different sources and in different temporal and spatial resolutions. Usually, irradiance data from these sources are recorded on the horizontal plane. In order to be used in PLR calculations, the irradiance has to be decomposed into its individual components and transposed to the plane-of-array irradiance of the PV system in question. Afterwards, the usage of these data is similar to the PLR calculation approach discussed in this study. A critical issue when using this type of data is the local component of the measurements.



Weather data recorded in different locations or averaged over a larger geographical grid, as it is the case for satellite data, are unlikely to capture the local climate conditions accurately in high resolution. The detection and removal of instances with non-logical power-irradiance pairs is thereby of utmost importance. Data alignment issues are especially pronounced in locations with complex topographies and prevalent micro-climates such as mountainous or coastal areas.

4.1.3. Nearby weather stations

Furthermore, especially in highly populated areas, nearby weather stations are possibly present. Again, if irradiance measurements are available, they are usually recorded on the horizontal plane and need to be decomposed and transposed to the plane-of-array. Depending on the proximity of the weather station and the PV system and also the density of nearby stations, the data of one station should be used or, if more than one station is available, then a weighted average between the available stations could be used. Unfortunately, there is no one-fits-all solution and it will always be a case-by-case decision. Furthermore, just like for PV system dedicated weather stations, a quality check in terms of calibration and cleaning procedure should be performed if possible.

4.1.4. Site adaptation techniques

Lastly, the application of site adaptation techniques may reduce the uncertainty in long-term irradiance time series coming from satellite and reanalysis sources. Here, the satellite dataset is combined with short-term high-quality measurements at a dedicated site coming from a relatively short measurement campaign. It is used to correct systematic errors and to reduce the bias in the original dataset [60].

4.2. Performance loss rate component break-down

As mentioned in [61], PLR is a metric that takes into account all types of performance loss that may exist within a PV system: reversible (i.e. the ones that can be recovered) and irreversible (i.e. the ones that cannot be recovered). Depending on the monitoring level, these losses include PV module and BOS degradation, soiling, snow, shading, fuse and tracking failures, etc (see figure 5). For example, maximum power point (MPP) data from continuous outdoor monitoring on the inverter-level can be used to estimate an overall system PLR whereas string- or module-level monitoring would represent string and module PLR, respectively. In monitoring applications, PLR is useful for an overall health state assessment of a PV plant (or multiple parts of it; if monitoring strings and/or modules). However, since this metric does not differentiate individual performance losses, it cannot provide any actionable insights for maintenance. Knowledge of individual performance losses is required for informing O&M activities, which requires a PLR breakdown. This can be achieved by using information from deploying specialized sensors, or by applying data analytic approaches, which can classify individual performance losses from PV time series [62]. The former option might be cost prohibitive whereas the latter is more reasonable since it can be applied on data that are already being continuously recorded.

Although rarely measured in commercial and utility power plants, continuous module IV curves may give attributes to what drives PLR. Furthermore, discreet indoor IV measurements from periodical PV module sampling and flash testing can provide information on module degradation rates, especially during the first years of deployment, where degradation might be highly nonlinear.

5. Summary and conclusions

In this work, an exhaustive summary and best practices for PLR calculation techniques has been presented. PLR is often assumed to be linear, or at least evaluated in linear terms, in order to be more understandable and accessible. That is why the majority of ongoing research until now has been focusing on the linearization of PLR. While this approach is very intriguing due to its apparent simplicity, it is known that the performance evolution of a PV system is usually nonlinear. In order to account for the nonlinearity in PV performance and PLR, nonlinear or change-point PLR calculation techniques have been developed recently. This paper covers both, the current state of the art of linear as well as nonlinear PLR calculation approaches. A linear evaluation still has its practical importance since it provides a high level view on the time-dependent performance drop of PV systems and therefore, it is very useful for quick health status checks and also for PV fleet evaluations.

In general, the PLR calculation is divided into the following five steps: (a) input data quality assessment, (b) data cleaning and filtering, (c) performance metric selection, (d) possible time series feature corrections and (e) the application of a statistical model to calculate the final PLR. Based on recent studies, it has been shown that there are no uniform approaches to reliably calculate linear PLR. Instead, the ensemble approach seems to provide the most reliable calculation techniques by taking the mean inlier estimates from different pipelines.

Nonlinear PLR methods divide a nonlinear performance trend into two or more data subsets, which are evaluated independently. The available nonlinear PLR calculation methodologies are still scarce and this topic is subject to active research and improvements. First promising results have been achieved using the Facebook Prophet regression model as well as piecewise regression for change-point detection in nonlinear performance trend functions.

A reliable calculation strategy of PLRs is important not only for health status checks of operating PV plants but also to increase the understanding of PV performance in general with respect to technology, installation conditions and prevailing climate stressors. The PLR provides a 'bird's-eye view' on performance impairing issues a PV system is exposed to including degradation of PV modules and BOS, soiling, shading, snow, tracking issues, BOS component failures, etc. A reliable PLR calculation, especially when accounting for nonlinearity, is the first step to subdivide performance reducing issues into its respective root-causes. Gathered knowledge through this type of studies will open possibilities to specifically tailor PV system projects based on local climate conditions to improve the reliability and lifetime of PV systems.

Another application area is in solar due diligence. The usage of nonlinear PLR calculation methodologies allows a time and cost efficient health analysis of PV systems. This could be used by potential buyers of operating PV systems to assess their prospective assets. The appearance of change-points in nonlinear time series might indicate performance anomalies which have taken place close to the change-points, which in turn reduces the amount of performance and maintenance data to be analyzed.

Data availability statement

The data generated and/or analysed during the current study are not publicly available for legal/ethical reasons but are available from the corresponding author on reasonable request.

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